Buwei He

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Education

University of South Carolina **Aug 2022 – Aug 2026**
Doctor of Philosophy (Ph.D.) in Biostatistics, GPA: 4.0/4.0 *Columbia, SC, U.S.*
Core course: Mathematical statistics, High dimensional data analysis, Advanced survival analysis, Linear statistical models, Advanced computational statistics, Biostatistical machine learning

Rutgers University **Aug 2019 – June 2021**
Master of Science (M.S.) in Biostatistics, GPA: 3.97/4.0 *New Brunswick, NJ, U.S.*
Core course: Statistical inference, Advanced regression methods, Biostatistics theory, Statistical learning, Categorical/Longitudinal data analysis, Clinical trial

Beijing Forestry University **Sep 2015 – June 2019**
Bachelor of Science (B.S.) in Biotechnology, GPA: 3.44/4.0 *Beijing, China*
Core course: College Mathematics/Physics/Chemistry/Linear Algebra, Experimental Design and Statistical Analysis, Botany, Zoology, Cell Biology, Molecular Biology, Proteomics, Microbiology, Biochemistry, Genetics

Experiences

University of South Carolina Big Data Health Science Center **Aug 2022 – Aug 2026**
Graduate Research Assistant advised by Dr. Jiajia Zhang *Columbia, SC, U.S.*

- Cleaned and preprocessed statewide EHR data from SCDHEC using deduplication, imputation, and outlier handling to build high-quality datasets.
- Collaborated with interdisciplinary medical researchers to plan studies, develop methods, and formulate statistical strategies.
- Applied traditional (e.g., survival analysis, longitudinal modeling) and advanced methods (e.g., high-dimensional regression, deep learning) to analyze complex health data and generate insights.
- Co-authored multiple peer-reviewed publications and conference presentations.

Fudan University Human Phenome Institute **Aug 2021 – June 2022**
Research Assistant advised by Dr. Sijia Wang *Shanghai, China*

- Cleaned and performed QC on large-scale WGS and Bead-chip data from a multi-ethnic Chinese cohort. Developed an automated R-based alignment/correction algorithm to enhance data consistency.
- Independently conducted GWAS using genomic and phenotypic data, applying multiple statistical models to identify significant SNP-trait associations.
- Contributed data analysis to the national key project "Survey of Anthropometric Characteristics of Chinese Ethnic Groups".

Rutgers School of Dental Medicine **Jan 2021 – June 2021**
Data Quality Analyst advised by Dr. Shou-en Lu *New Brunswick, NJ, U.S.*

- Led end-to-end data management for the Opioid Analgesic Reduction Study (OARS), collecting raw data from 9+ clinical centers, performing ETL, and integrating into REDCap.
- Automated daily/weekly clinical trial monitoring using SAS (PROC SQL, MACRO, ODS), tracking recruitment, retention, medication adherence, and SAEs.
- Conducted statistical analyses (e.g., descriptive and trend analysis); translated results into actionable insights to support enrollment strategy and risk management.

Selected Projects

Dissertation project 3: Machine learning-based proportional hazards mixture cure model with enhanced interpretability via shapley additive explanations **Dec 2025 – Mar 2026**

- Proposed a presmoothing-based estimation framework for a machine learning-based proportional hazards mixture cure model.
- Derived a flexible machine learning-based cure rate estimator using global survival stacking approach.
- Developed CureDeepSurv, a deep neural network for estimating survival distribution under the proportional hazards mixture cure model.

- Conducted simulation studies and real data applications to illustrate the superiority of the proposed method under high dimensional data and complex covariate relationships.
- Manuscript submitted to peer-reviewed journal, Biostatistics.

Dissertation project 2: Nonparametric cure rate estimation for the generalized odds rate mixture cure model with interval-censored data **May 2025 – Nov 2025**

- Proposed a novel method for estimating the cure rate in interval-censored data using the generalized Turnbull estimator.
- Updated likelihood functions and the corresponding EM algorithm for parameter estimation.
- Conducted an extensive simulation study demonstrating improved accuracy and robustness of cure rate estimation in small-sample, interval-censored settings.
- Delivered an abstract presentation at the Eastern North American Region (ENAR) 2026 conference.
- Manuscript submitted to peer-reviewed journal, Statistics in Medicine.

Dissertation project 1: Presmoothing-based cure rate estimation for the semiparametric accelerate failure time mixture cure model **Nov 2024 – May 2025**

- Proposed an improved EM optimization structure using kernel smoothed estimator for cure rate estimation.
- Derived the likelihood function and corresponding asymptotic properties by incorporating the kernel-smoothed estimator and the accelerated failure time-type latency within the framework of our newly proposed optimization structure.
- Conducted a simulation study demonstrating the superior accuracy of parameter estimates achieved by the proposed method in handling small sample sizes and high-dimensional covariates, compared to traditional approaches.
- Illustrated the superiority of the proposed method on a real-world HIV dataset from South Carolina Department of Public Health.
- Manuscript submitted to peer-reviewed journal, Computational Statistics & Data Analysis.

Suppression rate prediction using medical prescriptions deciphered by large language model among people with HIV **Oct 2024 – Jan 2025**

- Trained predictive models for HIV viral suppression rates by integrating structured data (lab results) and unstructured data (prescriptions, medical notes) using both traditional (regression-based approaches) and advanced (tree-based models, deep neural networks) methods.
- Queried medical prescriptions by SQL, applied BERT mdoel to extract informative features, and performed dimension reduction on the raw features by principal component analysis.
- Our findings demonstrated enhanced prediction accuracy by incorporating BERT-derived features into the deep neural network.

Impact of integrase strand transfer inhibitors on cardiovascular disease in people with HIV **Mar 2024 – Dec 2024**

- Developed an R algorithm to transform raw medical prescription data into standardized representations, reflecting longitudinal ART regimen patterns.
- Applied the time dependent Cox proportional-hazards model and the accelerated failure time model to assess the effect of integrase strand transfer inhibitors (INSTIs) on cardiovascular disease.
- Our findings highlighted the cardioprotective effects of INSTI-based ART regimens.
- Results were presented at the 2025 American Public Health Association (APHA) conference and published in Annals of Epidemiology.

A longitudinal analysis of the association between substance use disorders and sustained viral suppression among people with HIV in South Carolina **Jun 2023 – Mar 2024**

- Led and wrote the data analysis protocol in collaboration with PIs.
- Proposed a new longitudinal sustained viral suppression definition based on EHR dynamic viral load test results.
- Constructed a generalized linear mixed model (GLMM) to evaluate the effects of substance use disorders on sustained viral suppression among people with HIV.
- Revealed a significant risk associated with substance use disorders on sustained HIV viral suppression. Published the results in peer-reviewed journal, AIDS.

Skills

Professional Skills: Survival analysis, Mixture cure model, High-dimensional data analysis, Generalized linear mixed model, Genome-wide association study, Machine learning, Data visualization

Programming Skills: R (tidyverse, data.table, ggplot2, lme4, survival, smcure, ncvsurv, survML), Python (keras, tensorflow), SAS, Linux (plink), LaTeX

Practical Skills: Strong knowledge of statistics, Effective oral and written communication skills, Independent research and teamwork ability

Certifications: SAS Certified Specialist: Base Programming Using SAS 9.4

Honors & Awards

SmartState Center for Healthcare Quality 2024 – 2025 Junior Scholar **Sep 2024**
University of South Carolina, Columbia, SC, U.S.

Delta Omega Honorary Society in Public Health **May 2021**
Rutgers School of Public Health, New Brunswick, NJ, U.S.

Publications

[1] **Buwei He**, Bankole Olatosi, Jiajia Zhang, Sharon Weissman, Xiaoming Li, and Xueying Yang. "Impact of integrase strand transfer inhibitors on cardiovascular disease in people with HIV". en. In: *Annals of Epidemiology* (Nov. 2025). JCR: Q2, S1047279725003370. ISSN: 10472797. DOI: 10.1016/j.annepidem.2025.11.006.

[2] **Buwei He**, Shujie Chen, Xueying Yang, Bankole Olatosi, Sharon Weissman, Xiaoming Li, and Jiajia Zhang. "Association between substance use disorders and sustained viral suppression: a longitudinal analysis among people with HIV in South Carolina". In: *AIDS* (Nov. 2024). JCR: Q2. ISSN: 0269-9370, 1473-5571. DOI: 10.1097/QAD.0000000000004077.

[3] **Buwei He**, Matthew Lohman, Stephanie Aghamoosa, Tiejun Zhang, Bankole Olatosi, and Jiajia Zhang. "Aging-related diseases in the U.S. population: a data-driven approach using All of Us electronic health record data". In: *The Journals of Gerontology: Series A* 81.11 (Nov. 2026). JCR: Q1, glag231. ISSN: 1758-535X. DOI: 10.1093/gerona/glag231.

[4] Pieter Baker, Jiayang Xiao, Yunqing Ma, **Buwei He**, Bankole Olatosi, and Jiajia Zhang. "The role of Ryan White HIV/AIDS program services in sustaining viral suppression among people living with HIV: insights from South Carolina". In: *JAIDS Journal of Acquired Immune Deficiency Syndromes* (Jan. 2026). JCR: Q3. ISSN: 1525-4135, 1944-7884. DOI: 10.1097/QAI.0000000000003831.

[5] Xueying Yang, Ruilie Cai, **Buwei He**, Sharon Weissman, Bankole Olatosi, Xiaoming Li, and Jiajia Zhang. "Disruptions of healthcare visits and viral suppression for people living with HIV during the COVID-19 pandemic in the US". In: *BMC Infectious Diseases* 25.1 (Oct. 2025). JCR: Q2, p. 1295. ISSN: 1471-2334. DOI: 10.1186/s12879-025-11682-z.

[6] Atena Pasha, Shan Qiao, Jiajia Zhang, Ruilie Cai, **Buwei He**, Xueying Yang, Chen Liang, Sharon Weissman, and Xiaoming Li. "Changes in mental health care utilisation before and during the COVID-19 pandemic among people living with HIV in the USA: a retrospective cohort study using the All of US dataset". In: *BMJ Public Health* 3.2 (2025). JCR: Q3, e002173. ISSN: 2753-4294. DOI: 10.1136/bmjph-2024-002173.

[7] Fanghui Shi, Ruilie Cai, **Buwei He**, Xiaoming Li, Xueying Yang, Sharon Weissman, Bankole Olatosi, and Jiajia Zhang. "Sexual orientation, gender identity and virologic failure among people with HIV: a cohort study in All of US research program". In: *BMC Public Health* 24.1 (2024). JCR: Q1, p. 2091. DOI: 10.1186/s12889-024-19559-7.

[8] Xiaotong Yang, Bing Geng, Changxiong Zhu, Hongna Li, **Buwei He**, and Hui Guo. "Fermentation performance optimization in an ectopic fermentation system". In: *Bioresource Technology* 260 (2018). JCR: Q1, pp. 329–337. ISSN: 0960-8524. DOI: https://doi.org/10.1016/j.biortech.2018.03.101.

[9] Jihang Jiang, Xiawei Peng, Zhenxin Yan, **Buwei He**, Changxiong Zhu, Hui Guo, and Bing Geng. "Isolation and identification of potassium-solubilizing bacteria from rhizosphere soil of apple tree". In: *Chinese Journal of Agrometeorology* 38.11 (2017), pp. 738–748. DOI: 10.3969/j.issn.1000-6362.2017.11.006.

Conference Contributions

[1] Nonparametric approach for the generalized odds rate mixture cure model with interval-censored data
Joint Statistical Meetings (JSM), Boston, MA, U.S., 2026

[2] Presmoothing-based cure rate estimation for the semiparametric accelerated failure time mixture cure mode
Eastern North American Region (ENAR), Indianapolis, IN, U.S., 2026

[3] Impact of integrase strand transfer inhibitors on cardiovascular disease in people with HIV
American Public Health Association (APHA), Washington, D.C., U.S., 2025

[4] Sexual orientation, gender identity and virologic failure among people with HIV: a cohort study in all of US research program
National LGBTQ Health Conference, Atlanta, GA, U.S., 2024

[5] Association between substance use disorder and sustained viral suppression among people with HIV
Conference on Retroviruses and Opportunistic Infections (CROI), Denver, CO, U.S., 2024

[6] Utilizing All of Us data to examine the impact of COVID-19 pandemic on mental health among people living with HIV
12th International AIDS Society (IAS) Conference on HIV Science, Brisbane, Australia, 2023